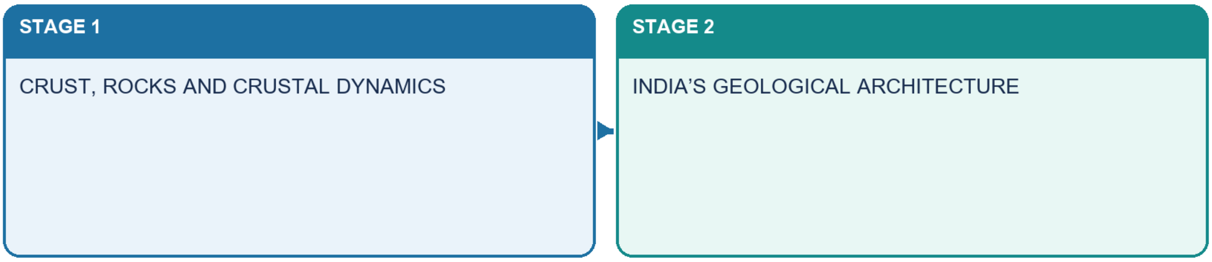
SOLVED PRACTICE WORKBOOK

Geography 02 - The Earth’s Crust, Rocks / India Geological Structure

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Two complete halves

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Geography 02 - The Earth’s Crust, Rocks / India Geological Structure

BASIC MCQS / REMEDIATION

33. Forty core diagnostic MCQs

Correct options rotate strictly **A -> B -> C -> D**, repeated ten times.

Q1. Which statement correctly separates a mineral from a rock?

- A. A mineral has characteristic composition and structure; a rock is an aggregate.
- B. Every rock is one crystal.
- C. Mineral and lithology are synonyms.
- D. A rock must contain metal.

Answer: A. A mineral has characteristic composition and structure; a rock is an aggregate.

Explanation: Minerals are constituents with characteristic chemistry/structure; rocks are natural aggregates and may contain several minerals or non-mineral material.

Q2. Which statement about crust and lithosphere is correct?

- A. They are identical everywhere.
- B. Lithosphere includes crust plus rigid uppermost mantle.
- C. Crust includes the whole asthenosphere.
- D. Lithosphere is only oceanic.

Answer: B. Lithosphere includes crust plus rigid uppermost mantle.

Explanation: The mechanical lithosphere extends below the compositional crust into rigid uppermost mantle.

Q3. Which process turns loose sediment into sedimentary rock?

- A. Melting
- B. Sublimation
- C. Compaction and cementation
- D. Complete metamorphism

Answer: C. Compaction and cementation

Explanation: Lithification is the compaction-and-cementation conversion of deposited sediment into rock.

Q4. Which statement best describes the rock cycle?

- A. It ends after metamorphism.
- B. Only igneous rock can weather.
- C. Sedimentary rock cannot melt.
- D. It contains multiple pathways rather than one fixed sequence.

Answer: D. It contains multiple pathways rather than one fixed sequence.

Explanation: Any family may weather, metamorphose or melt; the cycle is a network.

Q5. Slow cooling at depth most directly favours

- A. coarse crystals in intrusive rock.
- B. glassy basalt.
- C. mud cracks.
- D. slaty cleavage.

Answer: A. coarse crystals in intrusive rock.

Explanation: Slow cooling provides time for crystals to grow.

Q6. Which intrusive-extrusive pair is compositionally related?

- A. Basalt-granite
- B. Gabbro-basalt
- C. Shale-slate
- D. Diorite-rhyolite

Answer: B. Gabbro-basalt

Explanation: Gabbro and basalt are mafic partners with intrusive and extrusive occurrence respectively.

Q7. A porphyritic texture most safely indicates

- A. only chemical sedimentation.
- B. absence of crystals.
- C. two-stage cooling history.
- D. metamorphism without heat.

Answer: C. two-stage cooling history.

Explanation: Large early crystals followed by fine groundmass record changing cooling conditions.

Q8. Which feature is pyroclastic?

- A. A rounded river pebble
- B. A metamorphic foliation
- C. A chemical evaporite
- D. Fragmental volcanic ejecta consolidated into rock

Answer: D. Fragmental volcanic ejecta consolidated into rock

Explanation: Pyroclastic material consists of explosive volcanic fragments such as ash and lapilli.

Q9. Which is a clastic sedimentary rock?

- A. Sandstone
- B. Marble
- C. Basalt
- D. Gneiss

Answer: A. Sandstone

Explanation: Sandstone is made from lithified sand-sized clasts.

Q10. Loess is best described as

- A. lithified coral.
- B. wind-deposited fine sediment.
- C. an intrusive sill.
- D. a metamorphic band.

Answer: B. wind-deposited fine sediment.

Explanation: Loess is fine wind-laid silt and is not automatically lithified.

Q11. Which structure most directly indicates periodic drying of fine sediment?

- A. Vesicles
- B. Gneissic banding
- C. Mud cracks
- D. Batholiths

Answer: C. Mud cracks

Explanation: Polygonal mud cracks form when wet fine sediment dries and contracts.

Q12. Why is limestone classification context-dependent?

- A. It is always igneous.
- B. It never contains fossils.
- C. It cannot precipitate chemically.
- D. It may form through chemical or biogenic pathways.

Answer: D. It may form through chemical or biogenic pathways.

Explanation: Carbonate sediment can be precipitated chemically or built from biological material.

Q13. The parent rock before metamorphism is called

- A. protolith.
- B. pluton.
- C. regolith.
- D. craton.

Answer: A. protolith.

Explanation: Protolith is the immediate parent rock subjected to metamorphism.

Q14. Which pair is correctly matched?

- A. Bedding-metamorphic mineral alignment
- B. Schistosity-visible planar metamorphic fabric
- C. Vesicles-sedimentary fossils
- D. Cleavage-igneous crystal size only

Answer: B. Schistosity-visible planar metamorphic fabric

Explanation: Schistosity is a metamorphic alignment of visible platy minerals.

Q15. Which progression is a common increasing-grade teaching path?

- A. Granite->sandstone->coal
- B. Basalt->limestone->chalk
- C. Shale->slate->phyllite->schist->gneiss
- D. Conglomerate->obsidian->marble

Answer: C. Shale->slate->phyllite->schist->gneiss

Explanation: The sequence is a useful grade path for mud-rich protoliths, with acknowledged exceptions.

Q16. Which caution about graphite is correct?

- A. It is always igneous.
- B. It proves every coal seam metamorphosed.
- C. It is identical to coal.
- D. Carbon-rich material may form graphite, but coal->graphite is not universal.

Answer: D. Carbon-rich material may form graphite, but coal->graphite is not universal.

Explanation: Graphite origin depends on carbon-rich material and metamorphic history; it is not a universal coal transformation.

Q17. Porosity differs from permeability because

- A. porosity is void space while permeability is connected fluid transmission.
- B. permeability is mineral colour.
- C. porosity applies only to granite.
- D. they are exact synonyms.

Answer: A. porosity is void space while permeability is connected fluid transmission.

Explanation: Void quantity and connection are different properties.

Q18. Groundwater in crystalline hard rock commonly depends on

- A. abundant primary pores only.
- B. weathered mantles, joints and fractures.
- C. fossils.
- D. coal seams alone.

Answer: B. weathered mantles, joints and fractures.

Explanation: Secondary openings and weathering dominate many hard-rock aquifers.

Q19. Which is the strongest engineering-geology statement?

- A. All limestone foundations fail.
- B. Granite needs no testing.
- C. Site conditions and discontinuity orientation matter more than rock name alone.
- D. Basalt is uniformly permeable.

Answer: C. Site conditions and discontinuity orientation matter more than rock name alone.

Explanation: Engineering performance depends on discontinuities, orientation, weathering and load, requiring site investigation.

Q20. Which conclusion about black soil is safest?

- A. Every dark rock makes black soil.
- B. Climate is irrelevant.
- C. All basalt aquifers are identical.
- D. Basalt is important parent material, but relief, drainage, climate and time modify the soil.

Answer: D. Basalt is important parent material, but relief, drainage, climate and time modify the soil.

Explanation: Parent basalt matters, but pedogenesis also depends on climate, drainage, relief, organisms and time.

Q21. Why are continental-shelf edges used in drift reconstruction?

- A. They better approximate structural margins than changing shorelines.
- B. They show magnetic stripes.
- C. They are plate boundaries everywhere.
- D. They contain no sediment.

Answer: A. They better approximate structural margins than changing shorelines.

Explanation: Shelf edges are less altered than modern shorelines and better represent continental structure.

Q22. Wegener’s central weakness was

- A. lack of fossil evidence.
- B. an inadequate physical mechanism.
- C. absence of continental fit.
- D. denial of Pangaea.

Answer: B. an inadequate physical mechanism.

Explanation: The hypothesis lacked a physically adequate driver until ocean-floor evidence.

Q23. Palaeomagnetism helps continental reconstruction by recording

- A. modern rainfall.
- B. population density.
- C. former field orientation and latitude/rotation.
- D. river discharge.

Answer: C. former field orientation and latitude/rotation.

Explanation: Remanent magnetism can reconstruct past latitude, rotation and polarity.

Q24. Which is not a valid conclusion from matching fossils?

- A. Continents were once connected.
- B. Ocean barriers matter for land organisms.
- C. Several evidence types should be combined.
- D. Identical fossils prove continents never moved.

Answer: D. Identical fossils prove continents never moved.

Explanation: Fossils support former connection; they do not deny movement.

Q25. Symmetrical magnetic stripes are most directly associated with

- A. sea-floor spreading at ridges.
- B. river deposition.
- C. contact metamorphism.
- D. isostatic rebound only.

Answer: A. sea-floor spreading at ridges.

Explanation: Ridge-created crust records alternating polarity as matching bands.

Q26. Why does the Earth not expand under sea-floor spreading?

- A. Ridges stop after one eruption.
- B. Older lithosphere is consumed at subduction zones.
- C. Continents absorb all new crust.
- D. Transform faults destroy crust.

Answer: B. Older lithosphere is consumed at subduction zones.

Explanation: Subduction recycles older oceanic lithosphere and closes the mass budget.

Q27. Which setting produces an island arc?

- A. Continent-continent collision
- B. Transform boundary
- C. Ocean-ocean subduction
- D. Passive margin

Answer: C. Ocean-ocean subduction

Explanation: Ocean-ocean subduction generates a trench and island arc.

Q28. Which distinction is correct?

- A. Andes and Himalaya are identical volcanic arcs.
- B. Transform margins create crust.
- C. Continental collision destroys large volumes of buoyant crust by subduction.
- D. Andes reflect ocean-continent subduction; Himalaya reflect continent collision.

Answer: D. Andes reflect ocean-continent subduction; Himalaya reflect continent collision.

Explanation: The Andes are a continental volcanic arc above subduction; the Himalaya are a collision orogen.

Q29. The juvenile Wilson-cycle analogy is

- A. Red Sea/Gulf of Aden.
- B. Atlantic Ocean.
- C. Pacific Ocean.
- D. Himalayan suture.

Answer: A. Red Sea/Gulf of Aden.

Explanation: The Red Sea is the standard narrow-young-sea analogy.

Q30. The Airy concept emphasises

- A. only density variation with no roots.
- B. deeper crustal roots beneath higher relief.
- C. magnetic reversals.
- D. ocean expansion.

Answer: B. deeper crustal roots beneath higher relief.

Explanation: Airy columns have broadly similar density but different root depth.

Q31. Which chain best describes India’s national skeleton?

- A. Shield->volcanic arc->coral plain
- B. Plain->ocean ridge->shield
- C. Peninsular shield->collision Himalaya->foreland basin fill
- D. Himalaya->passive margin->mid-ocean ridge

Answer: C. Peninsular shield->collision Himalaya->foreland basin fill

Explanation: Collision loads and flexes the plate while erosion fills the foreland basin.

Q32. Which statement about a radiometric age is safest?

- A. It always dates sediment deposition.
- B. It replaces stratigraphy.
- C. One date fixes every event in a province.
- D. It dates a mineral event that requires geological interpretation.

Answer: D. It dates a mineral event that requires geological interpretation.

Explanation: Isotopic systems date events recorded by specific minerals, not an entire province automatically.

Q33. Which distinction is correct?

- A. Gondwana is a palaeocontinent; the Gondwana Supergroup is a fault-basin continental rock succession containing major coal measures.
- B. Both are the Deccan Trap.
- C. Both mean the Himalayan foreland.
- D. Gondwana Supergroup is an island arc.

Answer: A. Gondwana is a palaeocontinent; the Gondwana Supergroup is a fault-basin continental rock succession containing major coal measures.

Explanation: The palaeocontinent and Indian stratigraphic succession are distinct concepts; the latter preserves major coal-bearing continental basin successions.

Q34. Peninsular stability most safely means

- A. absence of faults.
- B. lower average recent deformation than active boundaries, not aseismic uniformity.
- C. complete flatness.
- D. only Archaean exposure.

Answer: B. lower average recent deformation than active boundaries, not aseismic uniformity.

Explanation: Relative stability is comparative and compatible with inherited-structure reactivation.

Q35. Cuddapah and Vindhyan are best described as

- A. young volcanic arcs.
- B. modern floodplains.
- C. Proterozoic sedimentary basin successions over older basement.
- D. oceanic trenches.

Answer: C. Proterozoic sedimentary basin successions over older basement.

Explanation: Both are old Proterozoic basin successions over older shield basement.

Q36. Which is not sufficient for petroleum occurrence?

- A. Source rock
- B. Reservoir and seal
- C. Appropriate trap and timing
- D. A sedimentary basin name by itself

Answer: D. A sedimentary basin name by itself

Explanation: A complete petroleum system is required; basin presence alone is insufficient.

Q37. Which Deccan Trap statement is correct?

- A. It is stacked extrusive basalt.
- B. It is a Gondwana coal measure.
- C. It is uniformly impermeable.
- D. It is entirely younger alluvium.

Answer: A. It is stacked extrusive basalt.

Explanation: Repeated fissure-fed basalt flows built the province.

Q38. The simplified south-to-north Himalayan order begins with

- A. Greater Himalaya.
- B. foreland/Siwalik before Lesser and Greater belts.
- C. Tethyan belt south of the plain.
- D. Deccan basalt.

Answer: B. foreland/Siwalik before Lesser and Greater belts.

Explanation: The plain and Siwalik lie south of Lesser, Greater and northern Tethyan/suture domains.

Q39. Which island pairing is correct?

- A. Lakshadweep-active subduction arc
- B. Andaman-stable craton
- C. Andaman-Nicobar-arc setting; Lakshadweep-coral atolls
- D. Both-uniform coral islands

Answer: C. Andaman-Nicobar-arc setting; Lakshadweep-coral atolls

Explanation: The island systems have contrasting tectonic and surface origins.

Q40. Which statement best controls resource determinism?

- A. Every shield is an ore body.
- B. Every Gondwana layer has coal.
- C. Every basin contains oil.
- D. Geology creates potential; processes, grade, preservation, technology and economics decide resources.

Answer: D. Geology creates potential; processes, grade, preservation, technology and economics decide resources.

Explanation: Resource association requires mineralising/depositional process, preservation and economic conversion.

34. Eight remedial MCQs

Rotation continues strictly **A -> B -> C -> D**, repeated twice.

Q41. A learner says ‘geological structure and physiography are synonyms’. What is the best correction?

- A. Structure concerns rock and tectonic architecture; physiography concerns broad surface divisions.
- B. Both terms mean relief only.
- C. Physiography concerns mineral chemistry only.
- D. Structure concerns climate zones.

Answer: A. Structure concerns rock and tectonic architecture; physiography concerns broad surface divisions.

Explanation: Geological structure is the concealed rock/tectonic framework; physiography is its broad surface expression.

Q42. A learner equates foliation with bedding. What is the correction?

- A. Both are cooling joints.
- B. Foliation is metamorphic fabric; bedding is depositional layering.
- C. Bedding forms only in marble.
- D. Foliation is a fossil.

Answer: B. Foliation is metamorphic fabric; bedding is depositional layering.

Explanation: The structures can look layered but record different processes.

Q43. A learner calls loess and moraine rocks automatically. What is the correction?

- A. Both are magmas.
- B. Both are metamorphic grades.
- C. They are deposits/sediments unless lithification is shown.
- D. Both are petroleum seals.

Answer: C. They are deposits/sediments unless lithification is shown.

Explanation: A deposit becomes rock only after lithification.

Q44. A learner says ‘new crust at ridges proves Earth expansion’. What is the correction?

- A. Ridges destroy continents.
- B. Magnetic stripes are unrelated.
- C. Oceanic crust never ages.
- D. Subduction balances ridge creation.

Answer: D. Subduction balances ridge creation.

Explanation: Ridge creation and trench destruction are paired.

Q45. A learner calls the Himalaya volcanic like the Andes. What is the correction?

- A. Himalaya is primarily a continent-collision fold-thrust orogen.
- B. Andes is a passive margin.
- C. Both lack earthquakes.
- D. Himalaya is an ocean ridge.

Answer: A. Himalaya is primarily a continent-collision fold-thrust orogen.

Explanation: Continent collision creates crustal thickening without an Andes-style volcanic arc.

Q46. A learner says ‘stable Peninsula means no earthquakes’. What is the correction?

- A. Only Himalaya has faults.
- B. Inherited rifts and faults can reactivate within a relatively stable plate interior.
- C. Kachchh is an island arc.

- D. Stability predicts dates.

Answer: B. Inherited rifts and faults can reactivate within a relatively stable plate interior.

Explanation: Kachchh, Koyna and Latur disprove the aseismic-peninsula shortcut.

Q47. Which correction best uses the Narmada-Tapi example without overclaiming?

- A. They are glacial valleys unrelated to structure.
- B. Every mapped linear feature is an active fault.
- C. They are structurally controlled west-flowing trough rivers, while present fault activity still needs field, geomorphic, geodetic and seismological evidence.
- D. Their courses prove that tectonic inheritance is absent.

Answer: C. They are structurally controlled west-flowing trough rivers, while present fault activity still needs field, geomorphic, geodetic and seismological evidence.

Explanation: Narmada and Tapi are structural-trough examples, but present fault activity still requires independent evidence.

Q48. A learner uses NCMM targets to claim a deposit exists. What is the correction?

- A. Mission policy replaces exploration.
- B. Every target is a reserve.
- C. All critical minerals occur in Deccan basalt.
- D. Mission objectives guide value chains but do not prove a specific deposit or reserve.

Answer: D. Mission objectives guide value chains but do not prove a specific deposit or reserve.

Explanation: Policy intent and exploration targets cannot substitute for geological evidence.

PYQS AND ANSWER PRACTICE

35. Audited PYQ ledger: nine genuinely relevant or adjacent routes

#	Year/paper	Route	Status and provenance
1	2022 GS-I Q4	Primary rocks	Direct; exact local official paper; UPSC publishes no model answer
2	2025 GS-I Q16	Continents/ocean basins	Direct; exact local official paper; independent model
3	2024 Prelims Q10	Fold/block mountains	Direct; local official Set-A paper and key **B**
4	2025 Prelims Q25	Drift evidence	Direct; local official Set-A paper and key **C**
5	2026 Prelims Q34	Peninsular Block	Direct; local official Set-A paper; **provisional key B**
6	2018 Prelims Q57	Magnetic reversal/early atmosphere	Direct; route retained; local official key unavailable; answer explicitly inferred
7	2023 Prelims Q9	Hill-location classification	Direct; exact local official paper; local official key unavailable; answer explicitly inferred
8	2022 GS-I Q6	Deccan Trap resources	Adjacent application; true owner Topic 31
9	2021 GS-I Q5	Gondwanaland/mineral base	Adjacent application; true owner Topic 31

[LIMIT] The audit count remains **nine**. No additional question is labelled a verified PYQ merely because it resembles the subject. The 2026 key remains provisional, and UPSC supplies no official model solutions for Mains.

36. UPSC GS-I 2022 Q4 - primary rocks

Exact local-paper wording: “Describe the characteristics and types of primary rocks.” (10 marks; answer in 150 words)

Independent model solution

[CLAIM] Primary rocks are igneous rocks formed when magma or lava cools and crystallises, providing much of the original mineral material later reworked into sedimentary and metamorphic rocks. [EVIDENCE] Intrusive or plutonic rocks such as granite, diorite and gabbro cool slowly below the surface and commonly develop coarse crystals. [ANALYSIS] Insulation delays heat loss, allowing crystal growth; later uplift and denudation expose them. [EVIDENCE] Extrusive rocks such as rhyolite, andesite and basalt cool rapidly at or near the surface and are commonly fine-grained; very rapid cooling can produce glass, while gas escape produces vesicles and explosive fragmentation produces pyroclastic rocks. [EVIDENCE] Composition ranges from felsic to mafic and helps explain colour, density, viscosity and weathering. India’s Deccan flood basalts and Peninsular granitic terrains illustrate the two modes. [LIMIT] “Primary” is genetic, not a claim that every igneous exposure is the oldest or most valuable. [VERDICT] Mode of occurrence, cooling texture and composition together provide the complete classification.

Why this earns marks: It follows “describe”, gives occurrence and composition, explains texture, uses Indian evidence and qualifies the word “primary”.

37. UPSC GS-I 2025 Q16 - continents and ocean basins

Exact local-paper wording: “Discuss how the changes in shape and sizes of continents and ocean basins of the planet take place due to tectonic movements of the crustal masses.” (15 marks; answer in 250 words)

Independent model solution

[CLAIM] Continents and ocean basins are temporary plate configurations, not permanent containers. [EVIDENCE] Shelf-edge fit, continuous rock belts, common fossils, palaeoclimate and palaeomagnetism establish former continental connections. [MECHANISM] Continental rifting stretches and thins lithosphere; continued divergence creates a narrow sea and then a mature ocean by sea-floor spreading. New basaltic crust records symmetrical magnetic stripes and becomes older and more sediment-covered away from the ridge. [MECHANISM] At convergent margins, oceanic lithosphere subducts, producing trenches and arcs and progressively reducing basin area. When buoyant continents meet, crust shortens and thickens instead of readily subducting, creating sutures, fold-thrust mountains and high plateaux. Transform faults offset boundaries without changing the crust budget. [EVIDENCE] East Africa, Red Sea, Atlantic, Pacific, Mediterranean and the Himalayan-Tethyan suture provide present-day stage analogies. India’s Gondwana separation, northward motion and Eurasian collision demonstrate simultaneous continental translation, ocean closure, mountain building and foreland formation. [LIMIT] The Wilson cycle is conceptual; rates vary and microplates, terranes and inherited structures complicate every margin. [VERDICT] Tectonics continually redistributes land and sea by creating, moving, deforming and recycling lithosphere.

Why this earns marks: It answers both shape and size, integrates evidence with mechanism, uses world and India examples and qualifies the ideal stage model.

38. UPSC Prelims 2024 Q10 - fold/block mountain matching

Exact local Set-A table:

Region	Mountain range	Type stated in question
Central Asia	Vosges	Fold mountain
Europe	Alps	Block mountain
North America	Appalachians	Fold mountain
South America	Andes	Fold mountain

Question: “In how many of the above rows is the given information correctly matched?” A. Only one · B. Only two · C. Only three · D. All four.

OFFICIAL LOCAL SET-A KEY: B.

Vosges is a block mountain, so row 1 is wrong. Alps is a fold mountain, so row 2 is wrong. Appalachians and Andes are fold systems, so rows 3 and 4 are correct. Exactly two rows are correct.

Why this earns marks: The elimination hinge is reversing the first two labels and distinguishing a rift-flank horst from fold belts.

39. UPSC Prelims 2025 Q25 - continental-drift evidence

Exact local Set-A wording:

“Which of the following are the evidences of the phenomenon of continental drift? I. The belt of ancient rocks from Brazil coast matches with those from Western Africa. II. The gold deposits of Ghana are derived from the Brazil plateau when the two continents lay side by side. III. The Gondwana system of sediments from India is known to have its counterparts in six different landmasses of the Southern Hemisphere.”

Options: A. I and III only · B. I and II only · C. I, II and III · D. II and III only.

OFFICIAL LOCAL SET-A KEY: C.

All three are accepted evidence routes: lithological continuity, mineral-source reconstruction and cross-Gondwana stratigraphic correlation.

Why this earns marks: Drift evidence is broader than coastline fit and includes rock, mineral, fossil, climatic and stratigraphic continuity.

40. UPSC Prelims 2026 Q34 - Peninsular Block

Exact local Set-A wording:

“Which of the following geographical features or phenomena is/are associated with the Peninsular Block of India?

1. Submergence of parts of the western coast due to tectonic activity
2. Presence of residual mountain ranges such as the Veliconda hills and Mahendragiri hills
3. Deep, V-shaped river valleys formed by fast-flowing rivers”

Options: A. 1 only · B. 1 and 2 · C. 2 and 3 · D. 3 only.

PROVISIONAL LOCAL SET-A KEY: B - NOT OFFICIALLY FINAL.

Statements 1 and 2 fit standard descriptions of the Peninsular Block. Statement 3 is not a defining general feature of the old denuded shield, even though local incision occurs.

Why this earns marks: It distinguishes isolated youthful valleys from the broad character of an ancient structurally varied block.

41. UPSC Prelims 2018 Q57 - magnetic reversal and early atmosphere

Wording retained by the repository route:

1. The Earth’s magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
3. When living organisms originated, they modified the early atmosphere of the Earth.

Options: A. 1 only · B. 2 and 3 only · C. 1 and 3 only · D. 1, 2 and 3.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: C (high confidence).

Statement 1 is correct in broad concept though reversals are irregular. Statement 2 is false: the early atmosphere was not oxygen-rich and carbon-dioxide-free. Statement 3 is correct because photosynthetic life altered atmospheric chemistry.

Why this earns marks: The implausible oxygen claim is the elimination hinge; palaeomagnetism and biological atmospheric modification are the valid concepts.

42. UPSC Prelims 2023 Q9 - Amarkantak, Biligirirangan and Seshachalam

Exact local-paper wording:

1. Amarkantak Hills are at the confluence of Vindhya and Sahyadri Ranges.
2. Biligirirangan Hills constitute the easternmost part of Satpura Range.
3. Seshachalam Hills constitute the southernmost part of Western Ghats.

Question: “How many of the statements given above are correct?” A. Only one · B. Only two · C. All three · D. None.

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: D (high confidence).

Amarkantak is linked with the Vindhya-Satpura meeting region, not Sahyadri. Biligirirangan lies in southern Karnataka at an Eastern-Western Ghats linkage, not the Satpura. Seshachalam belongs to the Eastern Ghats in Andhra Pradesh.

Why this earns marks: All three attach a real hill system to the wrong larger range.

43. Adjacent UPSC GS-I 2022 Q6 - Deccan Trap resources

Exact local-paper wording: “Discuss the natural resource potentials of ‘Deccan Trap’.” (10 marks; answer in 150 words)

Ownership: True owner Geography Topic 31; retained here as a bounded geology application.

Independent model solution

[CLAIM] The Deccan Trap’s resource potential arises from stacked basalt flows, weathering and structural discontinuities rather than basalt alone. [EVIDENCE] Basaltic parent material under seasonal conditions supports extensive black/regur soil tracts used for cotton, oilseeds and other crops. [MECHANISM] Clay-rich weathering products retain moisture, though drainage and climate modify productivity. [EVIDENCE] Vesicular flow tops, fractures, weathered contacts, dykes and intertrappean beds form local aquifers; compact interiors may transmit little water. [EVIDENCE] Basalt supplies building stone and aggregate, while step relief, escarpments and incised valleys create selected tourism, transport and hydropower opportunities. [LIMIT] Aquifers are heterogeneous, black soil can face waterlogging or shrink-swell constraints, and NCMM does not prove that all critical minerals occur in basalt. [VERDICT] The Deccan is a soil-water-construction-landscape resource system requiring site-specific management.

Why this earns marks: It derives several resources from flow architecture and weathering and avoids an exaggerated mineral list.

44. Adjacent UPSC GS-I 2021 Q5 - Gondwanaland and mineral economy

Exact local-paper wording: “Despite India being one of the countries of the Gondwanaland, its mining industry contributes much less to its Gross Domestic Product (GDP) in percentage. Discuss.” (10 marks; answer in 150 words)

Ownership: True owner Geography Topic 31; retained here to connect geology with economic conversion.

Independent model solution

[CLAIM] Gondwana inheritance helped create diverse Indian mineral settings, but geological endowment is not the same as realised economic value. [EVIDENCE] Ancient Peninsular mobile belts host iron, manganese, gold and base-metal provinces, while faulted Gondwana basins host major coalfields. [MECHANISM] Magmatism, sedimentation, metamorphism, hydrothermal activity, burial and preservation created deposits. [ANALYSIS] Mining’s contribution can remain limited because grade and depth vary; exploration is incomplete; transport, beneficiation and processing may lag; prices cycle; and land, ecological and community constraints appropriately raise the cost of extraction. [LIMIT] Gondwanaland cannot explain every ore or petroleum basin, and any GDP share requires a dated official source. [VERDICT] Policy must convert geological knowledge into responsible exploration, value addition, recycling and community-sensitive supply chains instead of equating extraction volume with development.

Why this earns marks: It separates endowment from economic conversion and combines geology, infrastructure, value addition and environmental-social qualification.

45. Four preserved map, chronology and cross-section drills

Drill 1 - Blank India structural map

Mark: Peninsular shield, Himalayan fold-thrust belt, Indo-Ganga-Brahmaputra foreland, Deccan Trap, Dharwar, Singhbhum, Cuddapah, Vindhyan, Damodar-Godavari Gondwana basins, Narmada-Son, Cambay, Andaman-Nicobar and Lakshadweep.

Solution logic: Use broad zones, not false boundaries. Keep the Himalayan arc north, alluvial foreland south of it, shield provinces across the Peninsula, basalt mainly west-central, Gondwana troughs east-/south-central, active arc in the Bay of Bengal and atolls in the Arabian Sea.

Drill 2 - South-to-north Himalayan section

Draw: alluvial foreland -> HFT/MFT -> Siwalik -> MBT -> Lesser Himalaya -> MCT -> Greater Himalayan crystalline -> Tethyan/Trans-Himalayan domain -> Indus-Tsangpo suture region.

Solution caution: Structures branch and vary laterally; the classroom order is conceptual.

Drill 3 - Broad chronology

Arrange: Archaean-Dharwar shield; Cuddapah-Vindhyan; Gondwana; Deccan flood basalt; Himalayan collision/foreland; Quaternary alluvium.

Solution: Archaean-Dharwar -> Cuddapah/Vindhyan -> Gondwana -> Deccan -> Himalayan collision/foreland -> Quaternary alluvium. Events overlap; this is an exam sequence, not a universal boundary table.

Drill 4 - Resource-process matrix

Match: shield/mobile belt-metallic ores; Gondwana rift basin-coal; complete sedimentary system-oil/gas; weathered/fractured hard rock-groundwater; basalt flow contacts-local aquifers; coastal sorting-heavy-mineral placers.

46. Six original solved Mains questions

Original 10-marker 1 - “The rock cycle is a network, not a one-way sequence.” Explain with geographical implications.

Model solution

[THESIS] The rock cycle links igneous, sedimentary and metamorphic rocks through several reversible pathways rather than a fixed classroom chain. [MECHANISM] Magma crystallises into igneous rock; any exposed rock can weather, erode and become sediment; burial, compaction and cementation produce sedimentary rock; heat, pressure and fluids metamorphose any suitable protolith; deep burial can melt rock; uplift exposes all families again. [EVIDENCE] Deccan basalt can weather to black-soil parent material, while Peninsular crystalline rocks yield fractured hard-rock aquifers; limestone can metamorphose into marble and sandstone into quartzite. [ANALYSIS] The network explains why rock age, present exposure and present land use are not identical. [LIMIT] Climate, relief, organisms, structure and time modify every weathering and soil outcome. [VERDICT] The cycle is therefore a material-transfer framework connecting deep Earth processes with soils, water, landforms and resources.

Why this earns marks: It explains pathways, uses Indian evidence, makes a geographic link and avoids the linear-cycle trap.

Original 10-marker 2 - Distinguish India’s geological structure from its physiographic divisions. Illustrate with two regions.

Model solution

[THESIS] Geological structure is the rock-and-tectonic architecture beneath India, while physiography is its broad surface expression. [EVIDENCE] The Peninsular Plateau is one physiographic division, but geologically it contains Dharwar, Bastar, Singhbhum and Bundelkhand blocks, Proterozoic basins, Gondwana rifts and Deccan basalt. [MECHANISM] Differential resistance, faulting and long denudation generate residual hills, scarps and structurally guided rivers. [EVIDENCE] The Northern Plain appears as level relief but is a flexural foreland depression filled by Himalayan and Peninsular alluvium. [MECHANISM] Orogenic loading created accommodation and river sediment built the plain. [LIMIT] Climate and rivers continue to reshape both provinces. [VERDICT] Physiography is the visible, modified expression of deeper geology, not a synonym for it.

Why this earns marks: It defines both terms, uses two regions, explains mechanisms and qualifies determinism.

Original 15-marker 1 - Analyse how rock properties control groundwater and engineering suitability.

Model solution

[THESIS] Groundwater and engineering performance depend on void connection, discontinuities, mineral fabric and weathering, not on rock name alone. [EVIDENCE] Sand and gravel aquifers transmit water through connected primary pores, whereas granite, gneiss and schist commonly depend on weathered mantles, joints and fractures. [ANALYSIS] Thus a crystalline terrain can be productive where fracture networks connect, yet nearly dry nearby. [EVIDENCE] In the Deccan Trap, vesicular flow tops and weathered contacts can store water while compact flow interiors and clay-filled fractures impede it. [EVIDENCE] Bedding, foliation and fault orientation influence slope failure, tunnelling and dam abutments; soluble limestone can develop cavities, while swelling clays create foundation movement. [MECHANISM] Water pressure reduces effective strength along discontinuities and weathering weakens mineral bonds. [LIMIT] Regional geology is only a screening tool; drilling, permeability tests, structural mapping and geophysics are required. [VERDICT] Applied geology converts rock properties into a site-specific hydro-mechanical assessment.

Why this earns marks: It covers aquifers and engineering, gives three named settings, explains mechanism and ends with investigation discipline.

Original 15-marker 2 - Explain how Peninsular evolution controls metallic minerals, coal and groundwater.

Model solution

[THESIS] Peninsular resource geography reflects successive crustal episodes rather than one uniformly mineral-rich shield. [EVIDENCE] Ancient Dharwar-Singhbhum-Aravalli belts experienced magmatism, metamorphism and hydrothermal activity, producing local iron, manganese, gold and base-metal provinces. [MECHANISM] Ore localisation required mineralising fluids, structures and preservation. [EVIDENCE] Later faulting formed Damodar, Son-Mahanadi and Godavari Gondwana basins where continental sediment, vegetation, burial and coalification created coal measures. [EVIDENCE] Crystalline rocks possess limited primary porosity, but weathered mantles, joints and faults store groundwater; Deccan basalt adds vesicular contacts and interflow zones. [ANALYSIS] Metallic ore, coal and aquifer maps differ because their processes differ. [LIMIT] Grade, recharge, technology, environmental constraints and economics determine usable resources. [VERDICT] The Peninsula is a layered system of ancient mineralised crust, younger sedimentary basins and structurally controlled aquifers.

Why this earns marks: All three demands receive named regions, distinct mechanisms and an economic-hydrological qualification.

Original 20-marker 1 - Trace the tectonic cycle from continental rifting to collision and assess its relevance to India.

Model solution

[THESIS] Rifting, ocean opening, subduction and collision form an idealised plate cycle that explains changing continents, ocean basins and India’s structural contrast. [MECHANISM] Extension domes and fractures continental lithosphere; continued stretching permits magma to create oceanic crust and a narrow sea. Sea-floor spreading enlarges the basin, while passive margins subside and accumulate sediment. Where subduction begins, trenches, arcs and inclined earthquake zones consume oceanic lithosphere. Basin closure brings continents together; buoyant crust shortens, thickens and uplifts into a suture and fold-thrust orogen. [EVIDENCE] East Africa, Red Sea, Atlantic, Pacific, Mediterranean and Himalaya are stage analogies, while magnetic stripes, ocean-floor ages and drift evidence test the sequence. [INDIA] The Indian block separated from Gondwana, moved north, closed Tethyan domains and collided progressively with Eurasia. Collision built the Himalayan-Tibetan orogen, flexed the Indian plate into a foreland basin and fed Siwalik and plain sediment. Earlier rifts also influenced western margins and Peninsular basins. [CONSEQUENCE] The architecture explains Himalayan seismicity, passive-margin petroleum opportunity, Gondwana coal basins and the shield-plain-mountain relief contrast. [LIMIT] The Wilson cycle is not a fixed destiny; terranes, microplates and inherited structures complicate it. [VERDICT] India is a particularly clear example of ancient shield inheritance being reorganised by an ongoing global plate cycle.

Why this earns marks: It traces the full mechanism, identifies evidence, applies it to India and delivers resource, relief and hazard significance with qualification.

Original 20-marker 2 - “India’s geological stability is relative, spatially uneven and socially consequential.” Analyse.

Model solution

[THESIS] Stability compares deformation regimes; it does not divide India into active and inactive halves. [EVIDENCE] The Himalaya and north-east form an active collision system with thrusting, crustal shortening and earthquakes. [MECHANISM] Continuing convergence stores strain, while thick alluvium in adjoining basins can amplify shaking. [EVIDENCE] Andaman-Nicobar lies on a subduction-related arc with earthquake, tsunami and volcanic contexts. [EVIDENCE] Peninsular cratons deform less on average, yet inherited rifts and faults produce intraplate risk in Kachchh and events such as Koyna, Latur and Jabalpur. [ANALYSIS] Far-field stress interacts with local weaknesses; “old” does not mean inert. [SOCIAL CONSEQUENCE] Dense mountain towns, vulnerable masonry, unstable slopes, dams, industrial corridors and soft-sediment cities convert geological processes into unequal risk. Resource consequences are also uneven: shield belts host metallic ores, Gondwana rifts coal, and sedimentary margins petroleum systems. [LIMIT] BIS zonation is a design generalisation rather than prediction, and geological association does not guarantee economic extraction. [VERDICT] Planning must replace the stable/unstable binary with microzonation, site investigation, resilient construction and responsible resource assessment.

Why this earns marks: It interprets every word of the quotation, compares four settings, links process to society/resources and ends with a qualified planning verdict.

Mains answer replication grid

Practice route	Non-negotiable opening	Visual/structure to reproduce	Qualification that prevents overclaim
10-marker: rock cycle	network of multiple pathways	compact cycle with uplift/exposure return	climate, structure and time modify outcomes
10-marker: geology versus physiography	define both terms before comparison	Peninsula and foreland examples	surface form is modified, not mechanically predetermined
15-marker: groundwater and engineering	properties and discontinuities outrank rock name	porosity-permeability and hard-rock fracture matrix	site investigation is indispensable
15-marker: Peninsular resources	successive crustal episodes created different systems	shield metals -> Gondwana coal -> fracture aquifers	association does not guarantee usable grade or yield
20-marker: tectonic cycle	idealised rift-to-suture sequence	Wilson-stage spine plus India application	stage analogies are not a fixed destiny
20-marker: relative stability	compare deformation regimes, not active/inactive halves	Himalaya-Andaman-Kachchh-Peninsula matrix	BIS zones guide design; they do not predict an event

Timed rewrite protocol: reproduce each thesis from memory, add one labelled mechanism visual, use at least two named Indian examples, insert one explicit **[LIMIT]** sentence and finish with a consequence-oriented verdict. A rewrite is complete only when the answer serves every directive word rather than merely listing facts.